

Aleksandr Bystritsky

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Personal information

I'm 2-year Bachelor of the Department of Applied Mathematics and Computer Science at the Moscow Institute of Physics and Technology. The main direction of development is system programming. I have skills in creating projects, debugging, and code optimization. GPA = 8.17/10.

Main project

Prog. Language	The project architecture is similar to a compiler toolchain. The source code (written in a custom language) is translated into an executable file, passing through the stages of an abstract syntax tree and an intermediate representation. At the frontend stage, parsing is implemented using recursive descent. The backend generates an executable file directly from the code's intermediate representation. C Graphviz
Function printf	An implementation of the original 'printf' function (C). Supports a variable number of arguments and many output specifiers. Implemented in assembly using a jump table. The result is compared with the original function. Nasm x86-64
HashTable	Optimization of a hash table (chaining method) with an elevated load factor. Identification of hotspots using a profiler and their optimization with SIMD instructions and inline assembly. Evaluation of the contribution of each optimization to the project's overall performance. C Perf SIMD
3D geometry	Detection of triangles that intersect in 3D space. Uses a broad-phase filtering of triangles based on a spatial grid (uniform grid). Work on the project is still ongoing, specifically 3D visualization using OpenGL. C++ OpenGL

Skills

Prog. languages	C/C++, x86 Assembler, Bash.
Other	Markdown, LaTeX.
Tools	Git, Make, CMake, Radare2.
Libraries	Graphviz, CSFML, GTK, OpenGL.
Languages	Russian (native), English (B1).
Soft skills	Concentration, teamwork, communication.

Philosophy of life

A journey of a thousand miles begins with a single step